

Apex Inventory

UI Requirement Specification

Inventory Items

Route: /inventory

Document Version: 1.0.0

Project: ReactExample (Apex Store Inventory Management)

1. Overview

The Inventory Items page is the primary stock management interface. Users can view all inventory records in a sortable table, search and filter items, adjust quantities inline, monitor expiry status, see upcoming order arrivals, and delete items.

2. Purpose

- Central registry of all store inventory with live quantity management.
- Enable quick stock adjustments (+/-) without navigating to Add Stock.
- Filter and search to find specific products efficiently.
- Surface expiry warnings and next replenishment arrival dates.

3. Page Header

- Title: 'Inventory Items'. Subtitle: 'View, search, and manage current store items'.
- Filtered Count panel: number of items matching current filters.
- Total Stock Units panel: sum of quantities for filtered items only.

4. Search & Filters

4.1 Search Input

Text search matches item name OR category (case-insensitive substring match).

4.2 Category Filter

Dropdown: 'All Categories' or specific category from GET /api/categories.

4.3 State Filter

Options: All States, Solid, Liquid.

4.4 Stock Status Filter

Options: All Statuses, In Stock (>5), Low Stock (1-5), Out of Stock (0).

4.5 Reset Filters

Button appears when any filter is active. Clears all four filter values.

5. Inventory Table Columns

Column	Description
Item Name	Product name, bold white text
Category	Category name string
State	Badge: solid (box icon) or liquid (droplet icon)
Stock Quantity	Minus/Plus buttons, count, status badge
Last Stock-In	Date from item.lastStockIn or 'N/A'
Next Arrival	Earliest pending order arrival or 'No orders placed'
Expiry Status	Date with expired/expiring soon warnings
Actions	Delete button (trash icon)

6. Quantity Adjustment Behavior

Minus button: decrements quantity by 1, minimum 0. Plus button: increments by 1. Calls PUT /api/items/:id/quantity with new quantity. Table reloads after update. If quantity increases above previous value, backend updates lastStockIn to today.

7. Expiry Logic

- Expired: expiry date is before today (midnight comparison). Shows red 'Expired!' label.
- Expiring soon: expiry within 0-7 days from today. Shows amber 'Expiring soon' label.
- No expiry: displays 'No expiry' in muted text when expiryDate is empty.

8. Next Arrival Computation

For each item, finds pending orders matching item name (case-insensitive). Sorts by expectedArrival ascending. Displays earliest date or 'No orders placed'.

9. Delete Item

Trash button triggers confirm dialog: 'Are you sure you want to delete this item from the inventory?' On confirm: DELETE /api/items/:id, then reload data.

10. Loading & Empty States

- Loading: 'Loading inventory data...' text shown while fetching.
- Empty filtered results: Layers icon, 'No inventory items found', suggestion to reset filters.

11. Data Dependencies

- GET /api/items - inventory list
- GET /api/categories - filter dropdown
- GET /api/orders - next arrival dates
- PUT /api/items/:id/quantity - quantity updates
- DELETE /api/items/:id - item removal

12. Technical Notes

- Component: src/pages/InventoryList.tsx